



Hair-thin fibre chips with full computing power

Scientists have created a flexible, hair-thin 'fibre chip' that integrates full computing power into thread-like fibres, enabling electronics that can be woven into clothing, tactile VR gloves, and advanced biomedical implants. Developed

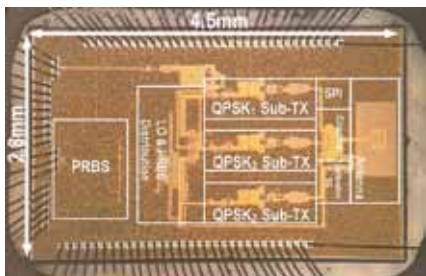


'Fiber Chip' shown at Fudan University in east China's Shanghai (Credit: Xinhua/Liu Ying)

by researchers at Fudan University in Shanghai, China, the fibre embeds processors, memory, and signal circuitry in a stretchable polymer and endures repeated bending and abrasion. Tests show it can host dense transistor arrays and process neural and digital signals, potentially merging computation with soft textiles for next-generation interactive wearables and brain-computer interfaces.

Ultrafast wireless transceiver matching fibre-optic speeds

Engineers at the University of California, Irvine, have developed a wireless transceiver that achieves data rates up to 120Gbps, matching the speeds of physical fibre-optic cables and laying groundwork for future 6G and FutureG wireless systems. Research conducted in UC Irvine's Samueli School of Engineering delivered a high-frequency (140GHz) silicon chip by merging



A new silicon chip wireless transmitter developed by electrical engineers at UC Irvine (Credit: Payam Heydari/UC Irvine)

digital and analogue processing to bypass traditional energy bottlenecks. The compact transceiver runs efficiently on standard 22nm fabrication, promising ultrafast wireless links in data centres, robotics, and edge computing.

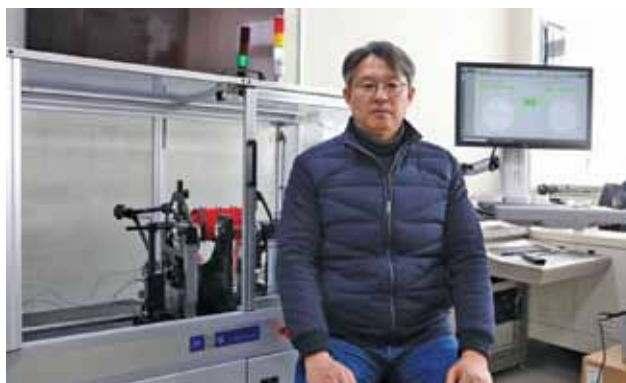
A cleaner approach to refrigeration

A zero-carbon refrigeration breakthrough using a novel dissolution barocaloric cooling method that integrates refrigerant and heat-transfer fluid into a single pumpable

liquid, potentially eliminating emissions tied to conventional vapour compression systems. Developed at the Chinese Academy of Sciences' Institute of Metal Research, the approach cycles pressure-driven salt dissolution and precipitation to achieve rapid cooling, dropping nearly 30K in 20 seconds and reaching speeds up to 54K in tests. The technique delivers high efficiency and heat transfer, offering promising cleaner cooling for homes, industry, and high-heat applications such as AI data centres.

AI powered precision measurement system

A new AI-powered precision measurement system has been unveiled to automate milligram-level imbalance detection and correction in industrial components, enhancing quality control for automotive, aerospace, energy, and defence



AI-based precision measurement technology (Credit: KIMM)

production lines. Developed at the Korea Institute of Machinery and Materials in South Korea with PNS Co. Ltd, the machine uses high-sensitivity vibration sensors, high-speed signal processing, and AI-based algorithms to filter noise, self-calibrate, and diagnose system health in real time. Its modular design supports direct integration into factory lines while improving repeatability and operational stability.

High-speed, long-range wireless laser communication

Indian startup olee.space has successfully demonstrated a 20km, 10Gbps wireless laser communication system that uses predominantly indigenous components to deliver high-capacity, long-range optical links for secure communication. The field-tested system, designed, manufactured and validated in India with about 85% local content, combines precision opto-mechanical subsystems and advanced stabilisation to maintain performance under dynamic atmospheric conditions. With carrier-grade availability and resistance to radio-frequency interference, the technology targets defence, strategic, and spectrum-constrained applications. The platform's architecture could be adapted for terrestrial, aerial, and future space-based networks.